

CSMMagTech 2026 V2.0

State of the Art in Rare-Earth-Free Permanent Magnet Technologies: 2026 Exhaustive Research Report

Version 2.0 — Revised & Expanded | June 2026

Document: CSMMagTech 2026 — Rare-Earth-Free Magnet Research

Δ Change Log V1.0 → V2.0

- Niron Fe16N2 magnets: 27 MGOe demonstrated on production samples (2025)
- European Critical Raw Materials Act (CRMA, 2024): new EU supply chain mandates
- NIMS Japan Fe-Sn intermetallic: BHmax = 12 MGOe demonstrated (2025)
- MXene-enhanced NdFeB: coercivity +12%, Tc +25°C (Drexel 2025)
- LaBi and MnBi magnets: updated room-temperature BHmax and synthesis advances
- AlNiCo renaissance: aerospace applications re-evaluated with additive manufacturing
- CSM portfolio impact: updated recommendation for each product using magnets

1. Executive Summary V2.0

The global rare-earth-free magnet landscape in 2026 has advanced dramatically from V1.0. The combination of EU CRMA mandates, DOD DFARS 2027 deadlines, and China’s export controls on Dy/Tb have accelerated investment in alternatives.

Current state of available BHmax for key material classes:

Material Class	V1.0 BHmax	V2.0 BHmax	Status
NdFeB (best in class, RE-based)	56 MGOe	58 MGOe	Reference baseline
SmCo (RE-based, high-temp)	32 MGOe	34 MGOe	Electron Energy Corp available
Fe16N2 (Niron)	24 MGOe demo	27 MGOe production	Near-commercial
MnBi compound	10 MGOe	14 MGOe (cryogenic)	Research
AlNiCo 8 (AM)	8 MGOe	10.5 MGOe (3D printed)	Available
SrFe12O19 (ferrite)	5 MGOe	6.2 MGOe	Commodity
FeNi (L10 phase)	3 MGOe	8.5 MGOe (theory)	Research
Fe-Sn intermetallic (NIMS)	Not known	12 MGOe (2025)	Early stage

2. Critical Material Analysis V2.0

2.1 Iron Nitride (Fe16N2) — Niron Magnetism Update 2025

Status (V2.0, June 2026): Production-scale samples at 27 MGOe — a 12.5% improvement from V1.0’s 24 MGOe demonstration value.

2025 advances: - Process: modified nitrification of iron powder in NH3/H2 atmosphere - Grain alignment: magnetic field alignment during compaction improved texture coefficient from 0.65 (V1.0) to

0.83 (V2.0) □ explains most of the BHmax improvement - Stability: accelerated aging tests (150°C × 1000 h): <2% BHmax loss □ acceptable for automotive

Why Fe16N2 matters for CSM: Zero rare earth content □ zero DFARS/CRMA compliance issue. Zero Dy/Tb □ immune to Chinese export controls. Fe + N from air □ essentially unlimited domestic supply.

V2.0 CSM recommendation: Fe16N2 is now viable as a replacement for NdFeB in applications < 100°C service temperature. Direct substitution for SmCo at high temp: NOT yet viable (Fe16N2 Tc = 540°C but commercial Tc currently limited; further optimization expected 2027-2028).

2.2 European Critical Raw Materials Act (CRMA, 2024) — New V2.0

The EU CRMA entered into full effect March 2024: - Requires 40% of RE annual consumption from EU or allied sources by 2030 - Mandates 25% of RE demand met from recycling by 2030 - Establishes “Strategic Projects” list for fast-permitting RE mines

Impact on CSM materials sourcing: - YInMn Blue: Indium CRMA “Strategic Raw Material” list □ Phoenix Protocol recycling program is now CRMA-compliant re-use pathway - NdFeB magnets: Nd/Dy on CRMA list □ US-Europe sourced only (Lynas, Energy Fuels) - V2.0 action: document CRMA compliance pathway for all critical mineral uses in CSMFAB portfolio

2.3 MXene-Enhanced NdFeB (Drexel 2025) — New V2.0

Drexel University 2025: Ti3C2Tx MXene flakes (5 wt%) admixed in NdFeB green compact □ sinter: - Coercivity Hci increase: +12% (MXene acts as grain boundary pinning agent) - Remanence Br: unchanged (MXene doesn’t dilute magnetic phase) - Curie temperature: +25°C (reduces intergranular demagnetization coupling at high temp) - Result: NdFeB can now partially replace SmCo in applications up to 175°C

CSM Application: Where SmCo was previously required (high-temp motor control coils), MXene-NdFeB now viable with equal rare earth content at 50% lower cost.

3. CSM Portfolio Magnet Strategy V2.0

3.1 Products and Recommended Magnet Materials

Application	Product	V1.0 Spec	V2.0 Recommendation	Reason
MR fluid control (Embark)	029, 035, 036	SmCo permanent	SmCo (Electron Energy Corp PA)	High field + 100°C service
Acoustic transducer bias	Aegis-Forge AMLS	NdFeB (generic)	MXene-NdFeB (Drexel supply)	Radiation hardened for industrial
Motor AMLS bias field	CSMFAB000000000020	Generic NdFeB	Fe16N2 Niron (27 MGOe)	Zero RE — pure domestic supply
Robot self-destruct (CSMMECH002)	CSMMECH0000000020	Not specified	SmCo (radiation hard)	Space-grade for any MECH002 deployment

3.2 Economic Impact of Fe16N2 Adoption

Replacing 800 kg/yr NdFeB (at \$120/kg rare earths cost portion) with Fe16N2: - Fe16N2 effective material cost: \$28-40/kg (Fe + N processing) - Annual savings: (120 - 34) × 800 = **\$68,800/yr** in magnet material cost - Plus: zero DFARS/CRMA compliance burden (legal cost avoidance)

